CSCI 3308 | Lab 8 Project Proposal

Tethr

A shared money and planning hub for friend groups

1. Team Number

Team Number: 6

2. Team Name

Team Name: Tethrers

3. Team Members

Name	GitHub Username	Email
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4. Application Name

Application Name: **t**ethr.

5. Application Description

Tethr is a web application that brings a friend group's shared money and shared plans into one place. Today, groups of college students, roommates, and travel companions juggle several tools at once: Venmo for payments, group chats for tracking who owes what, and separate calendars for coordinating plans. Tethr replaces that patchwork with persistent private groups called Circles, where members can log expenses, see balances, and plan events together.

Inside a Circle, users post an expense, record who paid, and tag the people who owe money. Expenses can be split equally, by custom dollar amounts, or by percentage, and every user's balance updates automatically. A dashboard shows the total a user owes and is owed across all of their Circles, while a People view combines balances with each individual friend across groups. Recurring charges such as rent, utilities, and streaming subscriptions can be set to repeat on a schedule, with reminders sent before they are due. Payments are settled outside the app through links to services like Venmo and then confirmed in Tethr.

What sets Tethr apart is that planning and spending live together. Each Circle has a shared calendar for birthdays, dinners, trips, and other events, with RSVP tracking, and expenses can be attached directly to an

event. A group planning a ski trip, for example, can create the event, track who is coming, split the Airbnb deposit and groceries, and settle up afterward without switching apps. The goal is a single, clear picture of what a group is doing and what everyone owes.

6. Audience

Ideal user: A college student, roughly 18 to 24, who lives with roommates and belongs to one or more active friend groups. They regularly share costs (rent, utilities, groceries, dinners, rideshares, trips) and coordinate plans through group chats. They are comfortable with mobile and web apps and already use payment apps like Venmo.

Problem we solve: Shared costs and shared plans are scattered across payment apps, group chats, notes, and calendars. Small debts get forgotten or become awkward to bring up, recurring bills are hard to keep track of, and nobody has one clear view of who owes whom or what the group has coming up. Tethr gives the group a single source of truth for both.

Secondary users: Roommates splitting household bills, people traveling together, and any small group that shares expenses on a recurring or one-off basis.

Usability and accessibility:

- Adding an expense takes only a few taps: name, amount, who paid, and who it is split between, with an equal split as the default.
- Balances are shown in plain language ("Emma owes you $45", "You owe Sophie $20") so users never have to do the math.
- Simple, consistent navigation (Home, Circles, Add, Calendar, Profile) with large, clearly labeled controls.
- Responsive layout that works on phones and laptops, readable color contrast, and text labels alongside icons and colors so meaning is not conveyed by color alone.
- Joining a Circle is as easy as following an invite link or entering a code.

7. Vision Statement

> **For** college students, roommates, and friend groups **who** constantly share costs and make plans together but track them across scattered apps and group chats. **Tethr is a** shared expense and planning app **that** keeps every group's balances, bills, and events in one place so friends always know who owes what and what's coming up. **Unlike** Splitwise, **our product** connects expense splitting with a shared group calendar, so the money and the plans behind it stay together.

Short version: *Keeping friends connected by making shared money and shared plans effortless.*

8. Version Control

GitHub Repository: https://github.com/Tethrers-Group-6/Tethr

The repository root will contain only the following:

- **TeamMeetingLogs** - weekly TA meeting minutes (decisions made, alternatives discussed, follow-up items and roles)
- **MilestoneSubmissions** - all course deliverables, including this proposal

- **ProjectSourceCode** - application source code and project documentation (includes .gitignore)
- **ReadMe.md** - project overview per the course project guide

9. Development Methodology

We are doing the Agile methodology with weekly scrum meetings.

10. Communication Plan

We are going to communicate via Slack and create a general channel, a PR request channel, a questions channel and an updates channel.

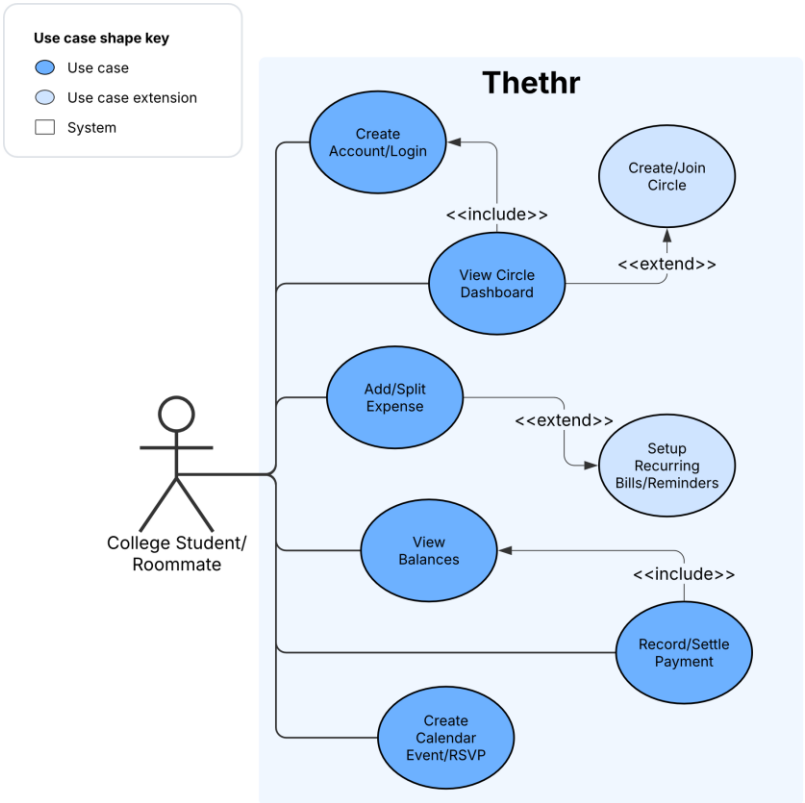
11. Meeting Plan

Team Meetings: Tuesdays 3-4 or longer as needed. Meetings will be on Zoom unless members are able to meet in person, in which we would meet in CSCI lounge.

Weekly TA Meeting: Thursdays between 3:15-5.

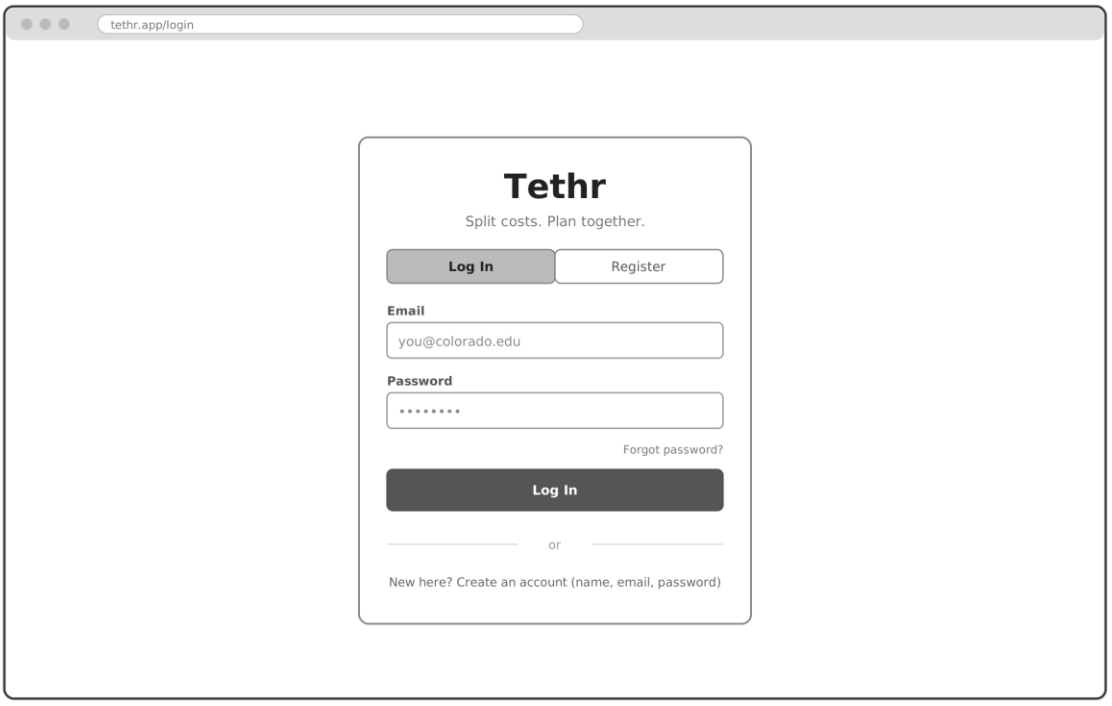
12. Use Case Diagram

13. Wireframes



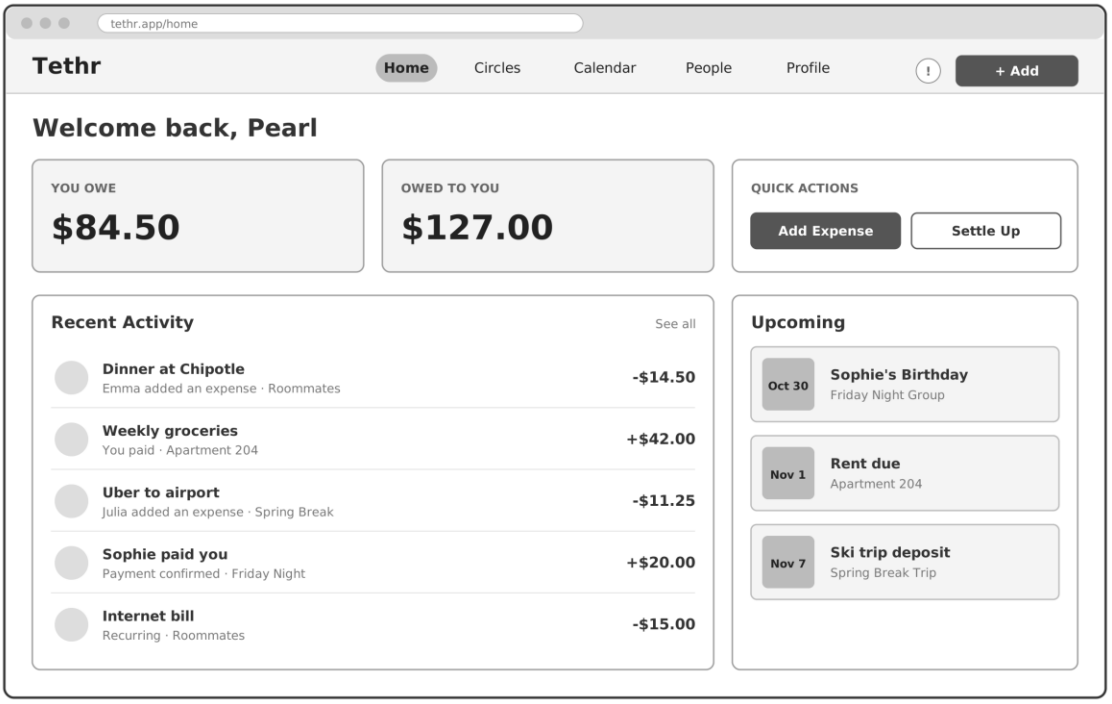
13.1 Login / Register

Entry point. Users log in with email and password or switch to the Register tab to create an account.



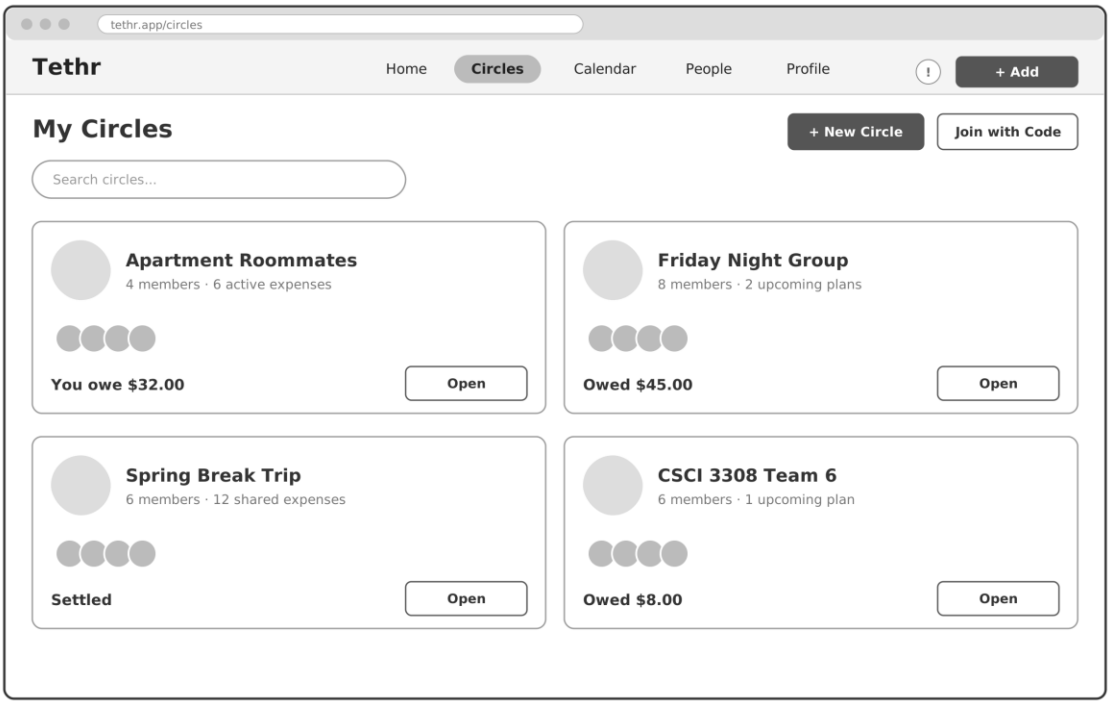
13.2 Home Dashboard

Totals owed and owed to the user across all circles, recent activity, quick actions, and upcoming events and bills.



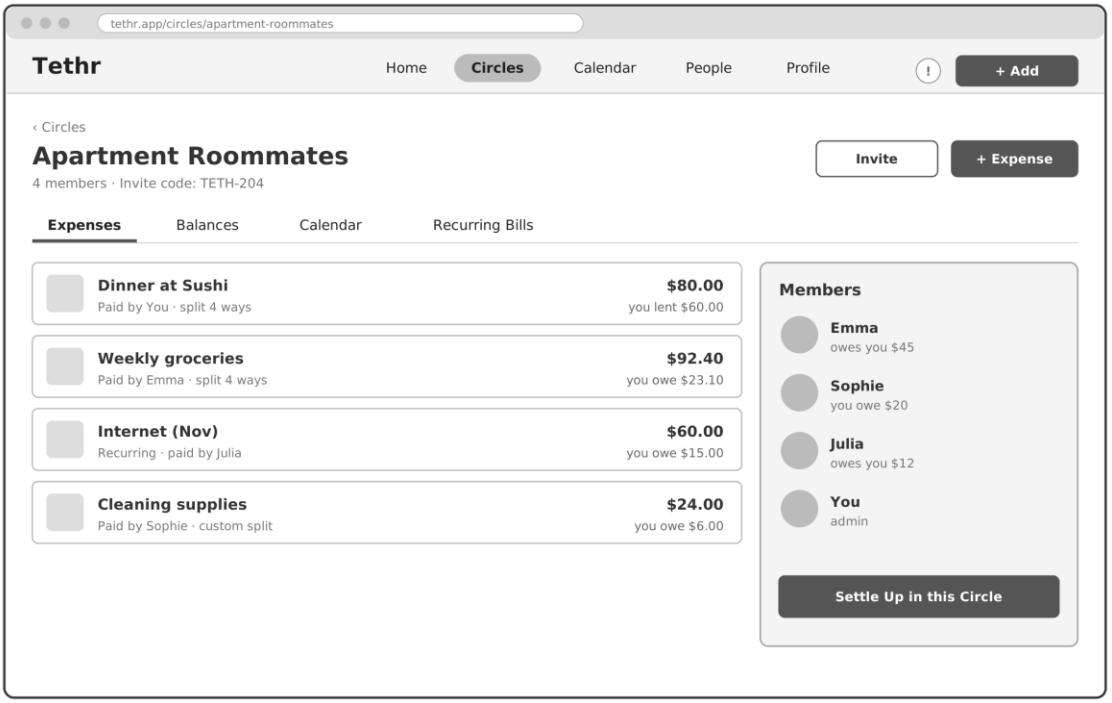
13.3 Circles

All of the user's friend groups with member count and current balance. Users can create a new circle or join one with an invite code.



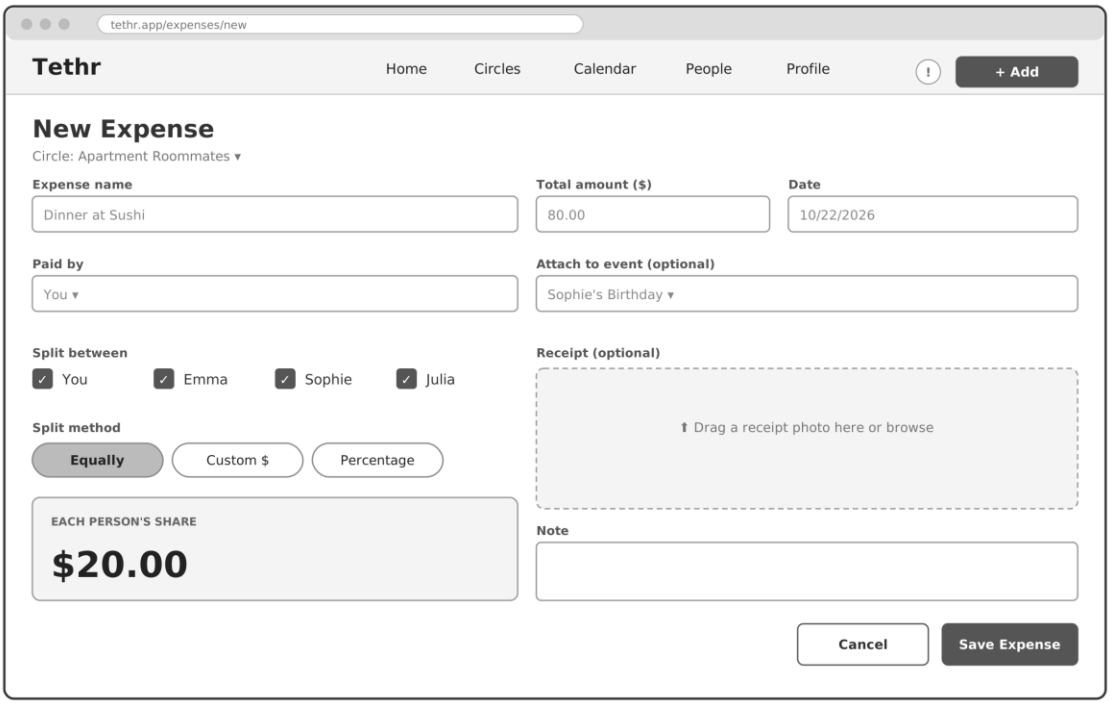
13.4 Circle Detail

A single circle's expense feed, with tabs for balances, the group calendar, and recurring bills. Members and settle-up live in the sidebar.



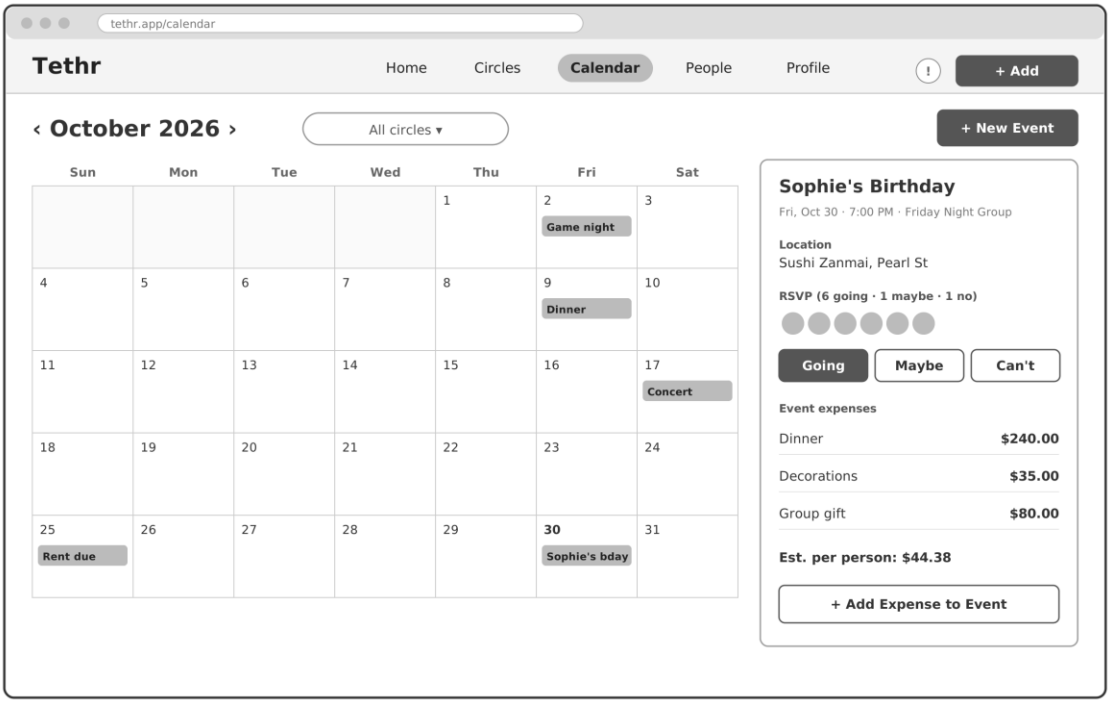
13.5 Add Expense

Create an expense: name, amount, who paid, who it is split between, and the split method. Optional receipt upload and event attachment.



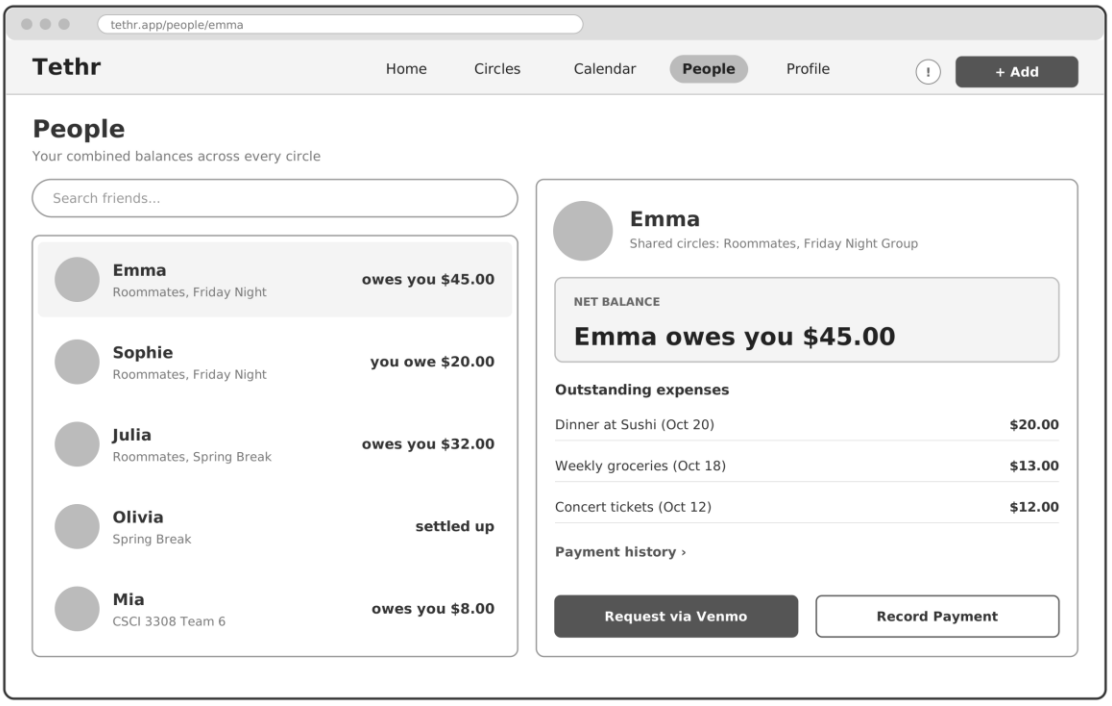
13.6 Calendar

Monthly view of events across circles. Selecting an event shows details, RSVP buttons, and the expenses attached to it.



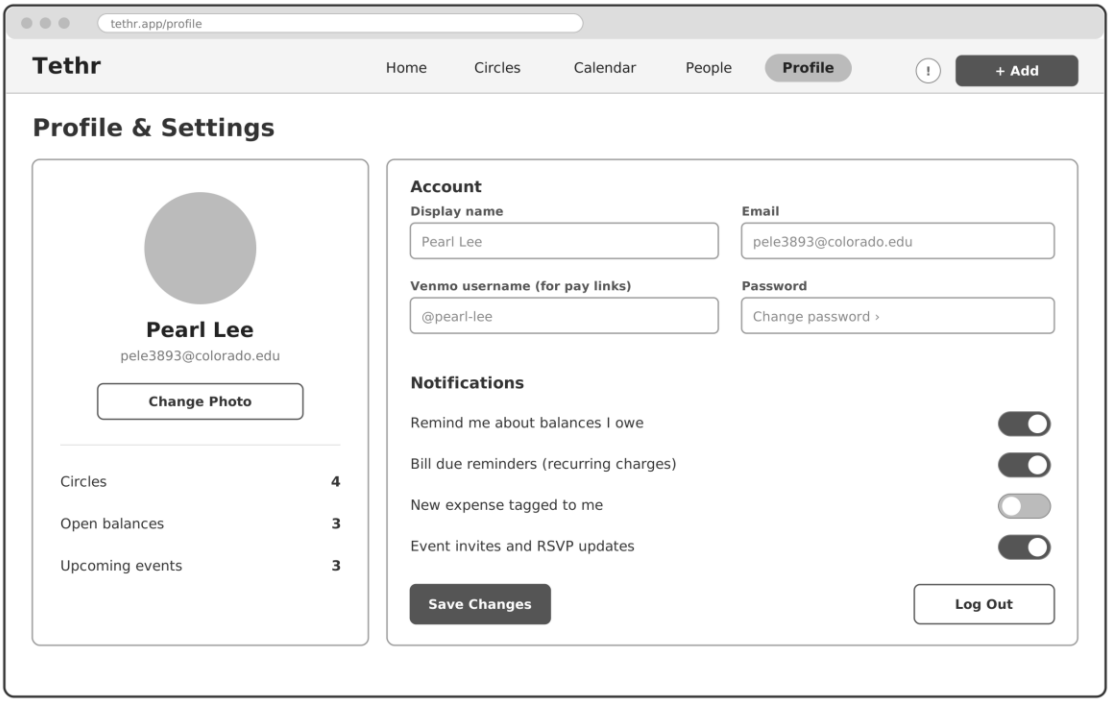
13.7 People / Balances

Combined balance with each friend across every circle, outstanding expenses, and options to request or record a payment.



13.8 Profile & Settings

Account details, Venmo username for payment links, notification preferences, and log out.



Extra Credit: Risk Analysis

Note: This section is our extra credit submission.

Risk	Severity Level	Mitigation Strategy
Incorrect balance math: rounding errors on uneven splits (e.g., $10 split 3 ways) or bugs in debt calculations	Medium	Double check all calculations in the testing stage of our application.
Privacy and social friction: users added to circles or tagged in expenses without agreeing	Low	We will require invite acceptance to join a circle, let users comment on or dispute an expense, and keep circle data visible only to members.
Exposing backend authentication information to users	High	Make sure only users have access to their own credentials, maybe implement partial masking, and don't return tokens to client
Scalability (large group events create multiple writes)	Low	Group expense writes will be batched instead of saved individually
Account login compromised	High	Account security, 2FA, re-authentication for larger transactions. System to detect new patterns in large transactions.